InCUBEation: a Series of Mission for Interplanetary Exploration using Small Satellite Platforms

Interplanetary Small Satellite Conference
28 May 2013, Caltech
Paper A.1.3

Alessandra Babuscia (MIT), Mary Knapp (MIT), Fernando M. Hicks (MIT), Mark Van de Loo (MIT), Benjamin Corbin (MIT), Sara Seager (MIT)

Outline

- Introduction
- CommCube-1
  - Mission goal and mission design
  - Mission status
- CommCube-2
  - Mission goal and mission design
  - Mission status
- ThrusterSat
  - Mission goal and mission design
  - Mission status
- SOLARA/SARA
  - Mission goal and mission design
  - Mission status
- Conclusion

Introduction

- MIT InCUBEation group is focused on 3 technological areas fundamental for interplanetary exploration using small platforms:
  - Communication →small satellites are limited in the amount of communication time and data rate (spacecraft and ground station constraints)
    - CommCube-1
    - CommCube-2
  - Propulsion → miniaturized propulsion systems are important to move beyond LEO, to perform orbital changes and to reach ambitious targets in the solar system.
    - ThrusterSat
  - Formation Flight → small spacecraft will inevitably be used in modular, fractionated formations. It is necessary to develop formation flight technologies and cooperative communication technologies.
    - SOLARA/SARA

CommCube-1: Mission Goal and Design

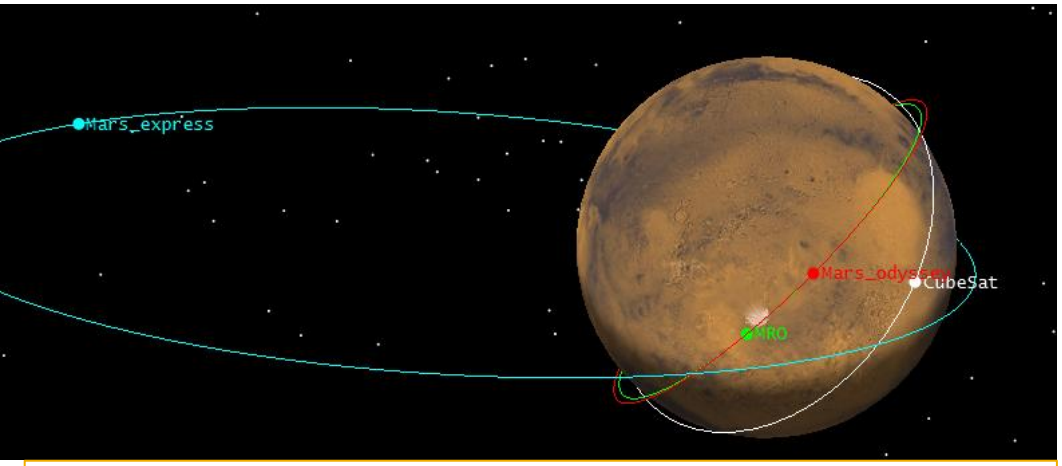
- CommCube-1 goals:
  - To measure the total communication throughput achievable by relaying on Globalstar phone network
  - To space qualify the Globalstar modem
- CommCube-1 is a 2U CubeSat.
- The communication system includes the Globalstar modem and a back-up UHF communication system.
- The MIT ground station is used for the UHF link
- Data downloaded using GlobalStar will be received through internet connection directly from the Globalstar gateways.
- The rest of the spacecraft (avionic, power) is designed using CubeSat COTS components.
- CommCube-1 is passively stabilized using permanent magnets and hysteresis material.

CommCube-1 CAD model [2]

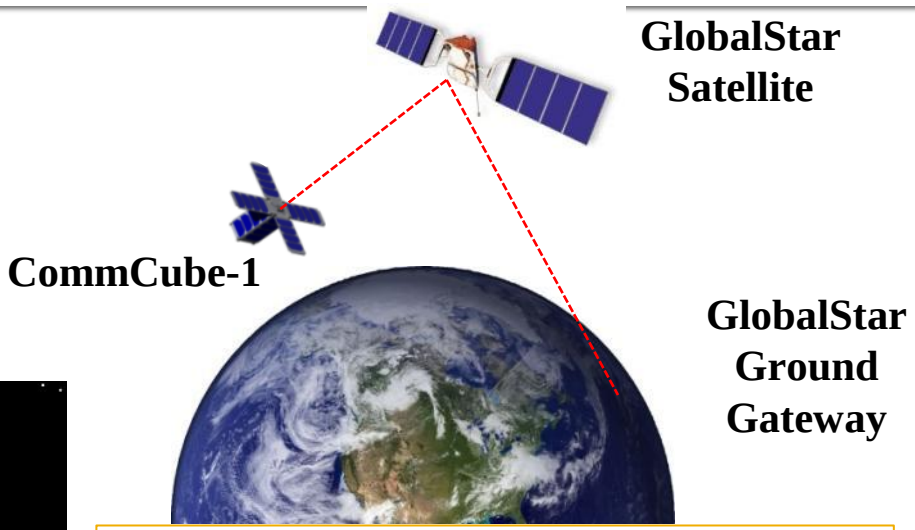
MIT UHF Ground station.

CommCube-1: Interplanetary Application and Mission Status

- **CommCube-1** will demonstrate the feasibility of high data volume transmission without traditional ground stations around the globe, leading to similar possibilities in the interplanetary scenario (example: Mars).



A simulation in STK reveals that the 3 Mars satellites could guarantee up to 86% coverage time for a CubeSat around Mars, which means the possibility of downloading up to 8 Gb of data per day (assuming a data rate of 115.2 Kbps which can be reached by state of the art CubeSat technology).



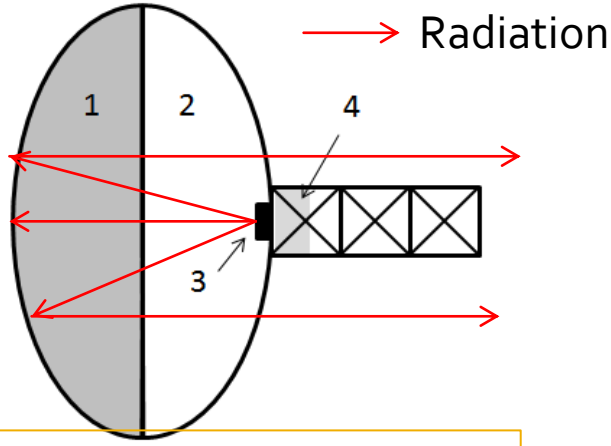
CommCube-1 communication strategy [2].

Mission Status:

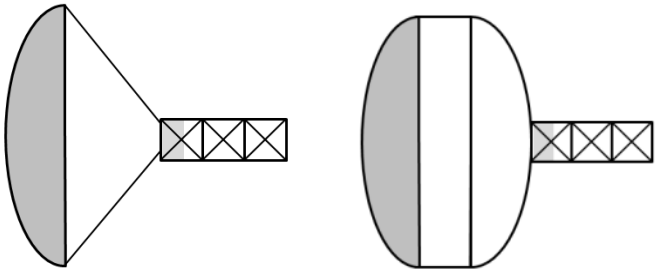
- CommCube-1 is in the design phase.
- Key avionic, power and communication components were tested and partially integrated.
- Coordination process with Globalstar for license is in progress.

CommCube-2: Mission Goal and Design

- CommCube-2 goals:
  - To test deployment and inflation of the inflatable antenna for CubeSat.
  - To measure the improvement in performance (gain) with respect to traditional patch antennas.
- The antenna is a 1-m diameter dish designed for the S-Band
- Inflatable volume with one side reflective, metalized Mylar (1), other side clear Mylar (2) with a patch antenna (3) at the focus.
- Stored in a volume (4) less than 1U in size.
- The antenna is passively inflated with a small amount of a sublimating powder.
- Building conical and cylindrical versions, trying to minimize deformation of the parabolic shape of the reflective section.
- The simulated gain is 16dB for the conical and 21 dB for the cylindrical configuration
- The CommCube-2 a 3U CubeSat (1U for the antenna, 1U for control, 1U for the rest of the electronic).



Concept design of the inflatable antenna [4].

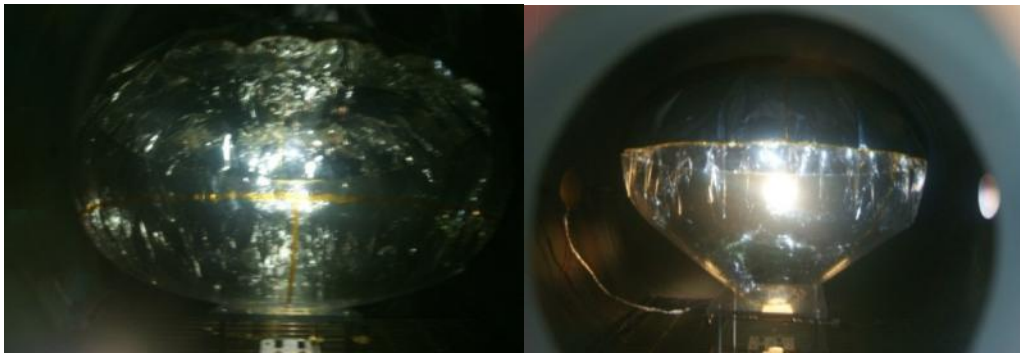


Conical (left) and cylindrical (right) configurations of the antenna [3].

CommCube-2: Mission Status

- Conical and cylindrical prototype were fabricated.
- The vacuum chamber tests showed
  - Successful inflation process, although residual air was still present inside the antenna
  - The shape of the reflective side was not perfectly parabolic
- The anechoic chamber tests showed losses due to leakage, attenuation caused by the polycarb supporting plate, loss of shape for the cylindrical configuration and poor performance on the patch antenna.
  - Data analysis is in progress
- The development of canister and deployment system will happen in Summer 2013

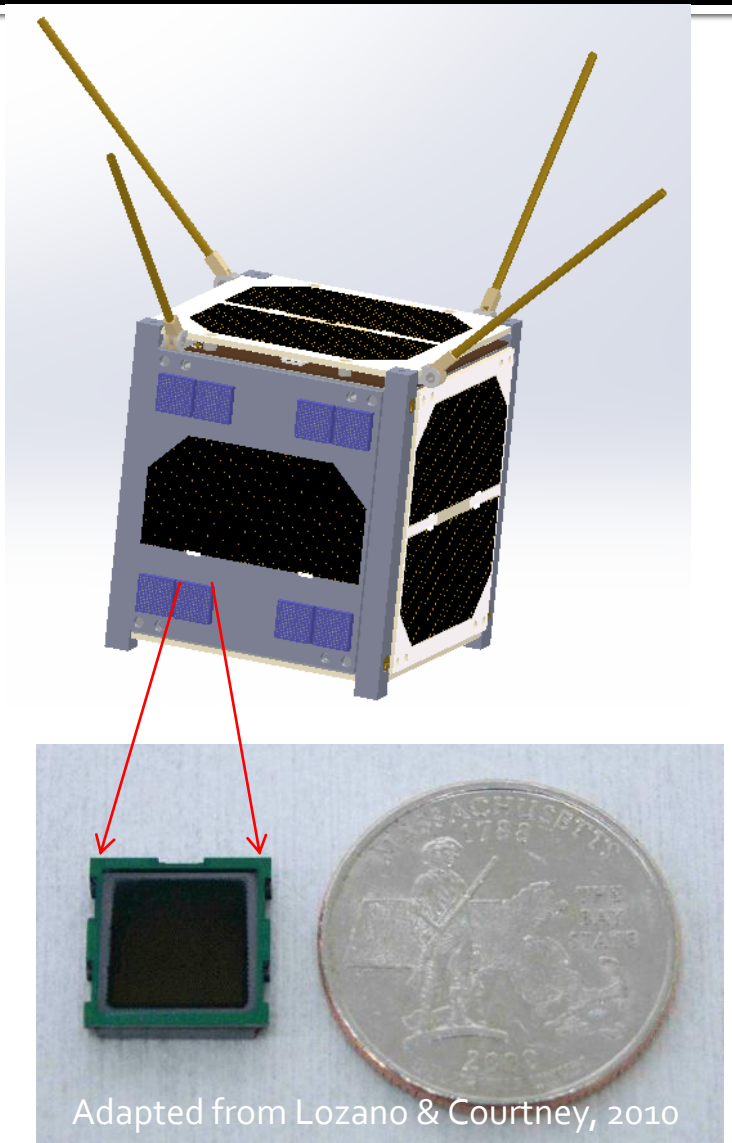
Cylindrical and conical prototypes



Antennas inflated in vacuum chamber

ThrusterSat: Mission Goal and Design

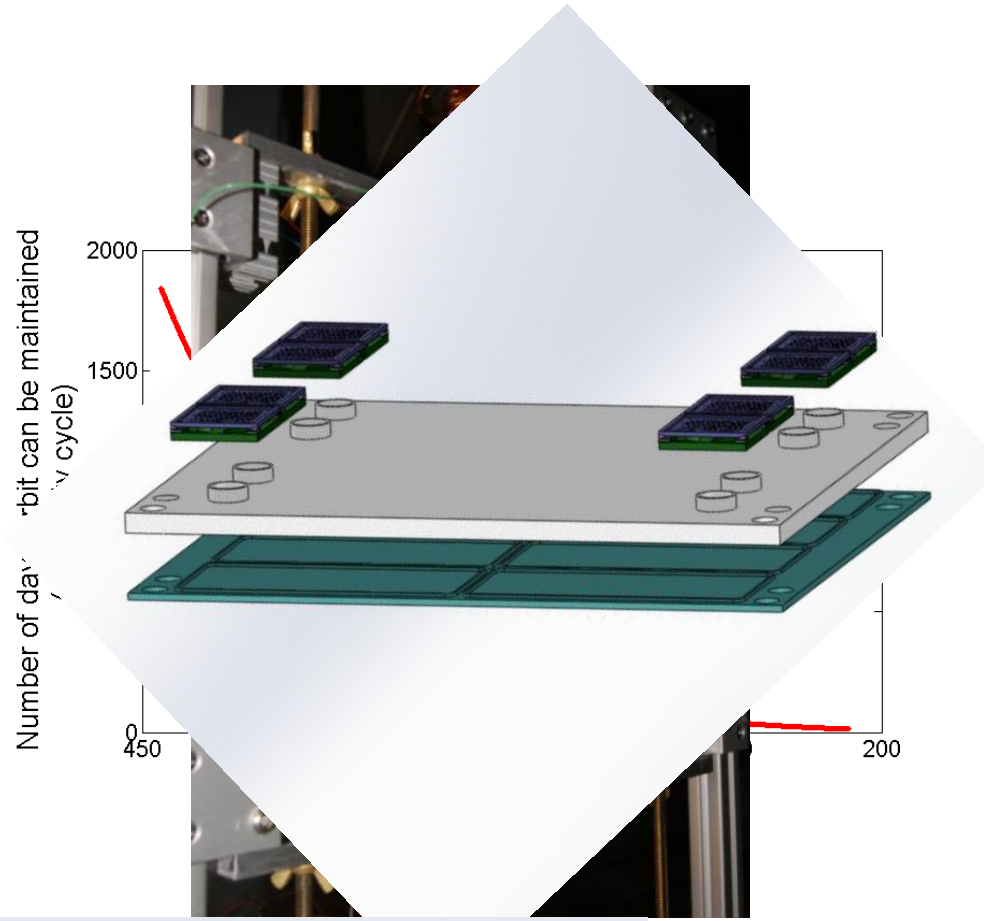
- TSat Goals:
  - Demonstrate micro-electrospray thrusters developed by MIT SPL in space environment
  - Demonstrate modular ADCS + delta-V system suitable for formation flight
- Isp > 3000, ~300 m/s delta-V, 2-axis attitude control
- 1U CubeSat deployed from ISS
- Novel in-panel fuel storage
- All components besides payload are COTS (GomSpace)
- UHF communication using MIT ground station
- GPS to track delta-V maneuvers

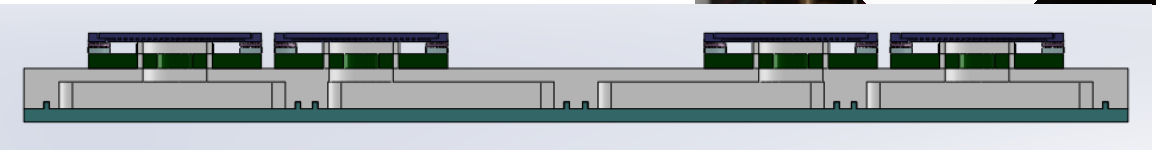


Adapted from Lozano & Courtney, 2010

ThrusterSat: Mission Status

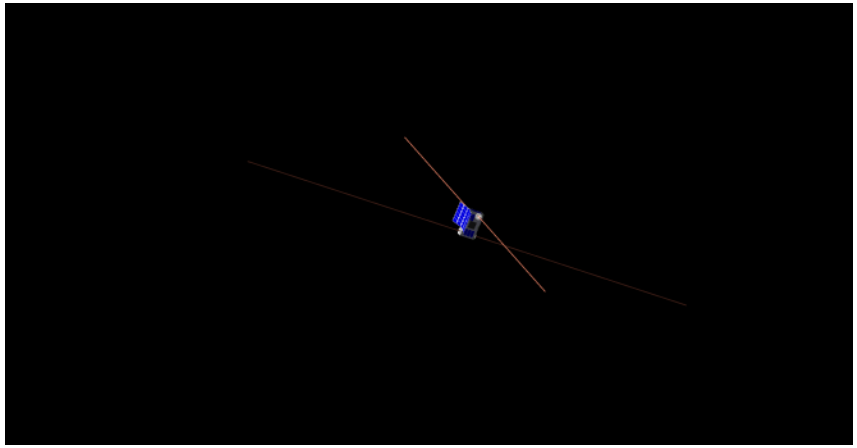
- Scheduled to launch mid-2014
- Magnetic balance used for system vacuum chamber testing
- Current work includes flat panel tank concept testing – fuel storage will be integrated into side solar/structural panels
- Small fuel mass fraction (1%), large delta-V capability
- Suitable for drag compensation in LEO, long-duration station-keeping at GEO, Lagrange points, or orbit modification maneuvers





SOLARA/SARA: Mission Goal and Design

- **S**olar **O**bserving **L**ow-frequency **A**rray for **R**adio **A**stronomy = ***SOLARA***
  - Addresses formation flight goal
  - Radio science below 30 MHz (ionospheric cut-off) – an unexplored region of the EM spectrum
  - Applications to heliophysics, exoplanets, cosmology and more
- **S**eparated **A**ntennas **R**econfigurable **A**rray = ***SARA***
  - Addresses communication goal
  - Phased antenna array composed of patch antennas on each spacecraft (S-band)



- 6U CubeSat x16
- Payload: Low-frequency radio science receiver + orthogonal dipole antennas (SOLARA)
  - FPGA-based
  - 1 Hz frequency tuning
  - Bandwidths from 1 kHz to 10 MHz
  - Optimized for 100 kHz to 10 MHz
- Distributed correlation among spacecraft
- Communication between spacecraft and to ground using S-band (SARA)

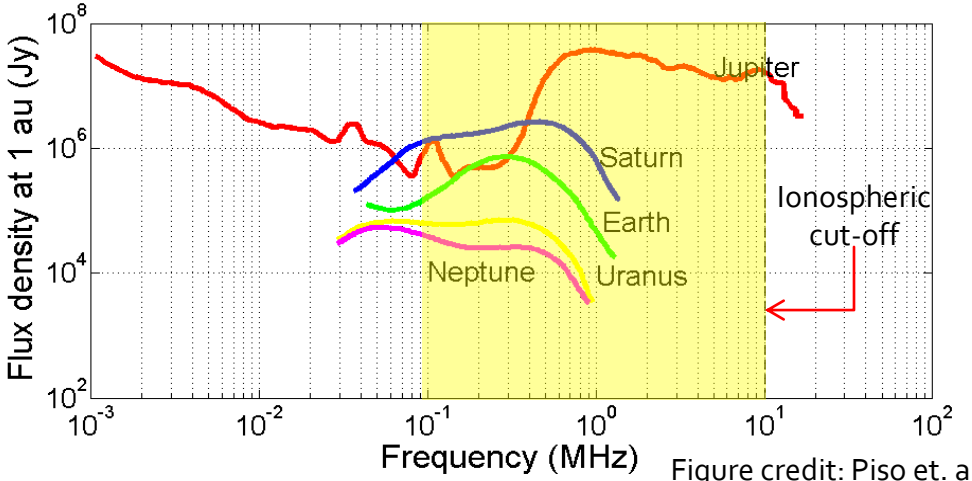
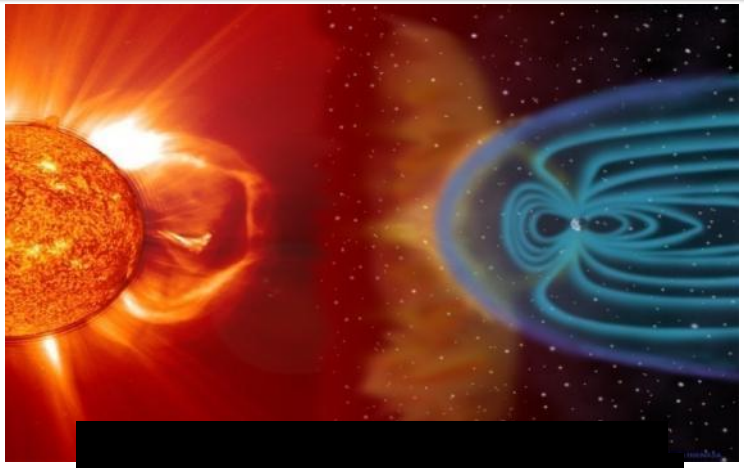
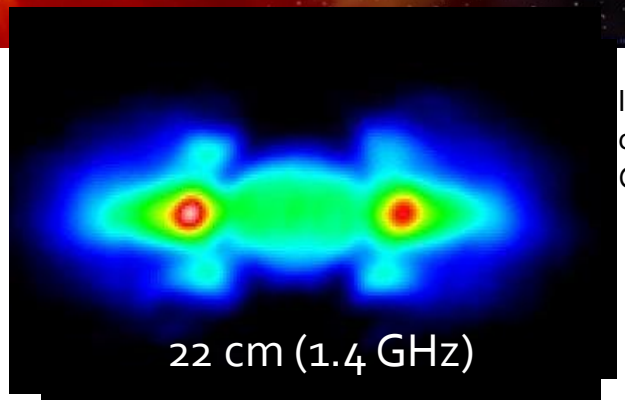


Figure credit: Piso et. al.

SOLARA/SARA: Mission Status

- Design and simulation underway for prototype formation – 3 spacecraft in Earth orbit
- Currently simulating some SOLARA features using a ground-based array
- Exciting science goals
  - Tracking CMEs (heliophysics)
  - Planetary and exoplanetary magnetospheric emission
  - 1st all-sky map at decent resolution below 30 MHz
- Pathfinder to future optical interferometry missions





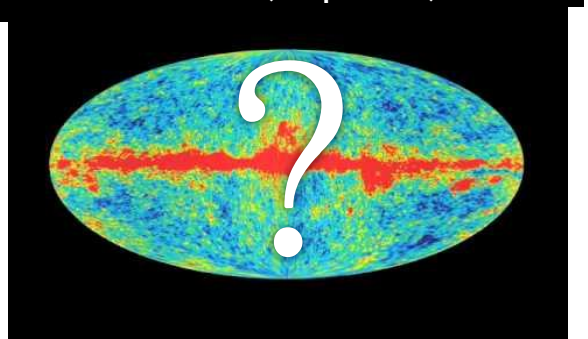


Image credit: CSIRO

Conclusion

- Four missions were summarized:
  - CommCube-1 → increasing data return for small spacecraft by using a network of relays.
  - CommCube-2 → flight demonstration of the inflatable antenna for CubeSat.
  - ThrusterSat → flight demonstration of the micro newton thruster technology.
  - SOLARA/SARA → array of CubeSats to perform radio astronomy observations while testing formation flight and cooperative communication techniques.

- Future efforts will be focused on finalizing the designs and on looking for possible launch opportunities for each of these missions over the next few years.

Massachusetts Institute of Technology

EAPS
Earth, Atmospheric and Planetary Sciences

Thank You!

Massachusetts Institute of Technology

EAPS
Earth, Atmospheric and Planetary Sciences

Questions?

References

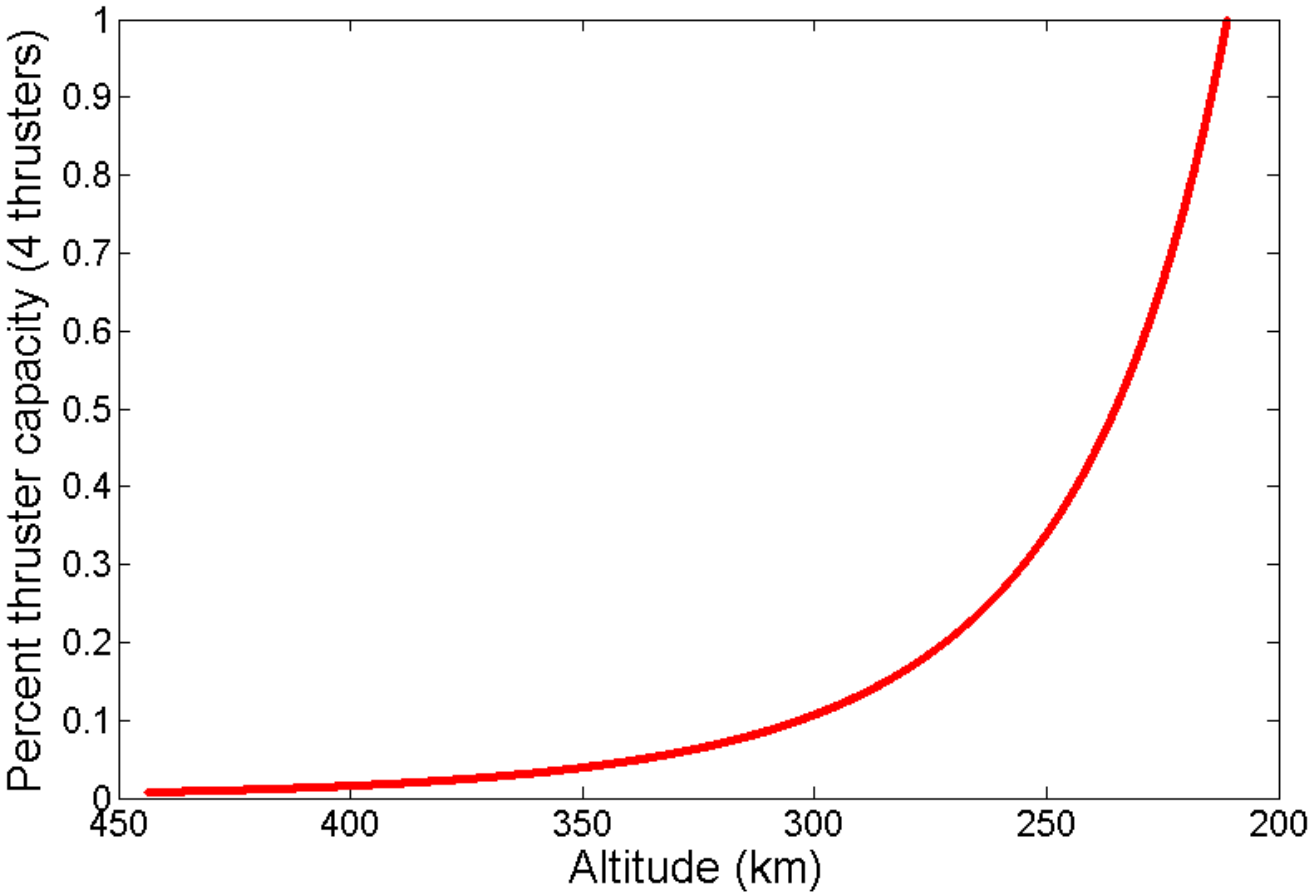
1. GomSpace company, http://gomspace.com/, accessed on 30th May 2013.
2. R. Jensen-Clem, M. Knapp, **A. Babuscia**, S. Seager, I. Sergeev, and M. Houston "CommCube-1: increase downlink volume through Globalstar network", **63rd International Astronautical Congress,** Naples, 2012.
3. A. Babuscia, B. Corbin, M. Knapp, R. Jensen-Clem, I. Sergeev, M. Van de Loo, and S. Seager, "CommCube 1 and 2: a CubeSat Series of Missions to Enhance Communication Capabilities for CubeSat", **IEEE Aerospace Conference,** Big Sky, March 2013.
4. **A. Babuscia**, M. Van de Loo, M. Knapp, R. Jensen-Clem, S. Seager, "Inflatable Antenna for CubeSat: Motivation for Development and Initial Trade Study", **Proceedings of Interplanetary CubeSat Workshop**, Boston, 2012.
5. **A. Babuscia**, B. Corbin, M. Knapp, R. Jensen-Clem, M. Van de Loo, S. Seager, "Inflatable Antenna for CubeSat: Motivation for Development and Antenna Design", **63rd International Astronautical Congress,** Naples, 2012.
6. M. Knapp, A. Babuscia, R. Jensen-Clem, F. Martel, "SOLARA/SARA: Solar Observing Low-frequency Array for Radio Astronomy/Separated Antennas Reconfigurable Array", 2nd Mission Idea Context, Nagoya, 2012 (**classified in the 1st place in the competition**).

Massachusetts Institute of Technology

EAPS
Earth, Atmospheric and Planetary Sciences

Back up

TSat Back-Up


Percent thruster capacity (4 thrusters)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
450
400
350
300
250
200
Altitude (km)